

The Maximal H^2 Calculus of the Bilateral Shift

Independent research note

August 13, 2026

Abstract

Ganesh–Bose–Rajagopalan leave as future work the unbounded operator $f(B)$ for $f \in H^2$, where B is the bilateral shift. We give a complete canonical realization. Under the Fourier model $\ell_2(\mathbb{Z}) \simeq L^2(\mathbb{T})$, $f(B)$ is the maximal multiplication operator by the L^2 boundary function of f . It is densely defined, closed, normal, and has an explicit adjoint, spectrum, and resolvent. Finitely supported sequences form a graph core, and on that core the Taylor power series converges in Hilbert norm and closes to the maximal operator. Bounded spectral truncations converge on the entire maximal domain. Thus both the domain and the required meanings of convergence are fixed.

Packet status

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Source	A. Ganesh, B. Bose, and A. Rajagopalan, <i>The Shift Operator Calculus for Stationary Time Series Analysis</i> , arXiv:2604.02336v2, conclusion, PDF page 7 [1].
Classification	Candidate full structural result, likely valid subject to human mathematical review.
Novelty	Low-to-moderate. The application and exact domain/core statement were not located, but the underlying Hardy-space and unbounded multiplication facts are classical.

6 Conclusion

We identified a gap in time series literature in the use of operator valued transfer functions $f(B)$, and addressed the gap with rigorous proofs of existence for the power series expressions denoted by $f(B)$ and $f(T)$, for $f \in H^\infty$.

We applied this rigorous operator calculus to unify the notion of stationary process invertibility based on the roots of the transfer function $f(z)$ with the notion of operator invertibility of $f(T)$. The article, in effect, rationalizes the vocabulary of time series analysis with known results in rigorous functional analysis. We also showed that $f(T)$ represents an isometry, and that $f(T) = T_f$, the Toeplitz operator.

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1 Proof intuition

The bilateral shift is the easiest possible unitary for unbounded functional calculus. Fourier transform sends it to multiplication by the coordinate z on $L^2(\mathbb{T})$. An H^2 function has an L^2 boundary value F , so the only natural maximal definition is multiplication by F on the vectors for which that product remains square integrable.

This model resolves the two ambiguities noted in the source. The domain is exactly a weighted L^2 space, making closedness, normality, the adjoint, and the resolvent transparent. For convergence, Taylor polynomials need not be controlled on every vector of the maximal domain. They do converge on the canonical core of finitely supported vectors, and closing that power-series operator gives the maximal realization. On the full domain, bounded spectral truncations provide unconditional graph convergence.

2 The maximal operator

Let m be normalized Haar measure on $\mathbb{T}$. On $\ell_2(\mathbb{Z})$, let $(e_k)_{k \in \mathbb{Z}}$ be the standard basis and

$$Be_k = e_{k+1}.$$

The Fourier unitary

$$U : \ell_2(\mathbb{Z}) \longrightarrow L^2(\mathbb{T}, m), \quad Ue_k(z) = z^k,$$

satisfies $UBU^{-1} = M_z$.

Let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \in H^2(\mathbb{D}),$$

and let $F \in L^2(\mathbb{T}, m)$ be its radial boundary function. Define

$$\text{Dom}(A_f) = \{x \in \ell_2(\mathbb{Z}) : FUx \in L^2(\mathbb{T})\}, \quad A_fx = U^{-1}(FUx). \quad (1)$$

Theorem 1. *The operator A_f in (1) has the following properties.*

(i) A_f is densely defined, closed, and normal, and

$$A_f^* = U^{-1}M_{\overline{F}}U, \quad \text{Dom}(A_f^*) = \text{Dom}(A_f).$$

It is bounded exactly when $f \in H^\infty$, in which case $\|A_f\| = \|f\|_{H^\infty}$.

(ii) Let $s_N(z) = \sum_{n=0}^N a_n z^n$. For every $x \in c_{00}(\mathbb{Z})$,

$$\sum_{n=0}^N a_n B^n x = U^{-1}(s_N Ux) \longrightarrow A_fx \quad \text{in } \ell_2(\mathbb{Z}).$$

Moreover $c_{00}(\mathbb{Z})$ is a graph core for A_f . Consequently the formal power-series operator on $c_{00}(\mathbb{Z})$ is closable and its closure is exactly A_f .

(iii) If $F_j = F \mathbf{1}_{\{|F| \leq j\}}$, then

$$A_{F_j} := U^{-1}M_{F_j}U \in \mathcal{B}(\ell_2(\mathbb{Z}))$$

and $A_{F_j}x \rightarrow A_fx$ for every $x \in \text{Dom}(A_f)$.

(iv) *The resolvent and spectrum are*

$$\lambda \in \rho(A_f) \iff \operatorname{ess\,inf}_{\zeta \in \mathbb{T}} |F(\zeta) - \lambda| > 0, \quad \sigma(A_f) = \operatorname{ess\,ran} F.$$

When $\lambda \in \rho(A_f)$,

$$(A_f - \lambda I)^{-1} = U^{-1} M_{1/(F-\lambda)} U.$$

(v) *Every e_k lies in $\operatorname{Dom}(A_f)$, and*

$$A_f e_k = \sum_{n=0}^{\infty} a_n e_{k+n}, \quad \|A_f e_k\| = \|f\|_{H^2}.$$

Proof. Under U , the operator A_f is the maximal multiplication operator M_F . Its graph norm is

$$\|h\|_{M_F}^2 = \|h\|_2^2 + \|Fh\|_2^2 = \int_{\mathbb{T}} |h|^2 (1 + |F|^2) dm.$$

The measure $d\mu = (1 + |F|^2) dm$ is finite. Bounded functions lie in $\operatorname{Dom}(M_F)$ and are dense in $L^2(\mu)$, so the domain is dense. Completeness of $L^2(\mu)$ proves that M_F is closed.

For g, h in the appropriate domains,

$$\langle M_F g, h \rangle = \langle g, M_{\overline{F}} h \rangle.$$

The standard maximal-domain argument gives $M_F^* = M_{\overline{F}}$. Since $\operatorname{Dom}(M_F) = \operatorname{Dom}(M_{\overline{F}})$ and $M_F M_{\overline{F}} = M_{\overline{F}} M_F = M_{|F|^2}$, the operator is normal. A multiplication operator is bounded exactly when its multiplier is essentially bounded, with norm $\|F\|_{\infty}$. The Hardy boundary theorem identifies $F \in L^{\infty}$ with $f \in H^{\infty}$ and preserves the norm [2].

Now take $x \in c_{00}(\mathbb{Z})$. Then $p = Ux$ is a bounded trigonometric polynomial. Taylor convergence in H^2 gives

$$\|(s_N - F)p\|_2 \leq \|p\|_{\infty} \|s_N - F\|_2 \longrightarrow 0,$$

which proves the asserted power-series convergence.

It remains to prove that this small domain is a core. Trigonometric polynomials are dense in $L^2(\mu)$: continuous functions are dense for the finite regular Borel measure μ , and the trigonometric polynomials are uniformly dense in $C(\mathbb{T})$. Since the $L^2(\mu)$ -norm is precisely the graph norm of M_F , the inverse Fourier image $c_{00}(\mathbb{Z})$ is graph dense in $\operatorname{Dom}(A_f)$. The closure claim follows.

For the spectral truncations and $h \in \operatorname{Dom}(M_F)$,

$$\|(F_j - F)h\|_2^2 = \int_{\{|F| > j\}} |Fh|^2 dm \longrightarrow 0.$$

This proves full-domain convergence in (iii).

Finally, $M_F - \lambda I = M_{F-\lambda}$. It has an everywhere-defined bounded inverse exactly when $1/(F - \lambda) \in L^{\infty}$, equivalently when the displayed essential infimum is positive. This is precisely the complement of the essential range, proving (iv). For (v), $Ue_k = z^k$, so

$$Fz^k = \sum_{n=0}^{\infty} a_n z^{n+k} \quad \text{in } L^2(\mathbb{T}),$$

and Parseval gives the norm identity. □

3 Consequences and convergence boundary

Corollary 1. *If $f \neq 0$, then A_f is injective and has dense range. It is surjective with a bounded inverse in the closed-operator sense exactly when $1/F \in L^\infty(\mathbb{T})$; then*

$$A_f^{-1} = U^{-1}M_{1/F}U.$$

Proof. A nonzero H^2 function has nonzero boundary values almost everywhere [2]. Hence both $\ker M_F$ and $\ker M_F^* = \ker M_F^*$ vanish, proving injectivity and dense range. The final assertion is Theorem 1(iv) with $\lambda = 0$. $\square$

The theorem also marks a necessary limitation. From $f \in H^2$ alone, Taylor partial sums are guaranteed to converge on the graph core, not on every vector of $\text{Dom}(A_f)$. Indeed, as maps from the graph-norm domain to $L^2(\mathbb{T})$, their norms are

$$\text{ess sup}_{\mathbb{T}} \frac{|s_N|}{\sqrt{1 + |F|^2}}.$$

Whole-domain pointwise convergence would force these quantities to be uniformly bounded by the uniform boundedness principle. No such estimate follows from H^2 membership. The bounded spectral truncations in Theorem 1(iii) give the unconditional full-domain replacement.

For stationary-process notation, part (v) is the essential concrete fact: the white-noise coordinate vectors are in the domain, and the moving-average series converges in mean-square geometry exactly because $(a_n) \in \ell_2$. Thus the unbounded operator is not merely formal even though it cannot act on all of $\ell_2(\mathbb{Z})$.

4 Verification, limitations, and novelty

The proof is unitary-model independent up to the harmless convention $Be_k = e_{k+1}$; reversing the shift replaces $F(z)$ by $F(\bar{z})$. All domain equalities use maximal multiplication domains. The graph-core step was separately checked through the finite weighted measure $(1 + |F|^2)dm$, and the resolvent formula includes the required verification that $F/(F - \lambda) = 1 + \lambda/(F - \lambda)$ is bounded whenever the inverse multiplier is bounded.

A bounded search on August 13, 2026 covered the run indexes, arXiv:2604.02336, the exact future-work sentence, and combinations of H^2 , unbounded $f(B)$, bilateral shift, maximal multiplication operator, graph core, and unbounded Toeplitz operator. It found standard accounts of unbounded multiplication operators and a broader unbounded-Toeplitz literature, but no later paper explicitly answering this source item and no exact bilateral-shift domain/core result tied to it. The result should therefore be read as a self-contained application and synthesis of classical tools, with low-to-moderate novelty confidence. It does not claim universal Taylor convergence on the maximal domain or analyze stochastic questions beyond the Hilbert-space moving-average action.

References

- [1] A. Ganesh, B. Bose, and A. Rajagopalan, *The Shift Operator Calculus for Stationary Time Series Analysis*, arXiv:2604.02336v2 (2026).
- [2] P. L. Duren, *Theory of H^p Spaces*, Academic Press, 1970.
- [3] J. B. Conway, *A Course in Functional Analysis*, Springer, 1997.